

Gait Differences in the Real World and Virtual Reality: The Effect of Prior Virtual Reality Experience

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ABSTRACT

Walking is one of the most influential parts of virtual reality applications, such as training, health care, and education. Prior research has established that people's gait behavior change when walking in virtual reality (VR), as compared to the real world. However, very little research has evaluated how people's gait parameters change over time as they gain more experience with VR, and whether this reduced the difference in gait parameters between the real and virtual worlds. We hypothesize that the more experience people have in VR, the less difference there will be between VR and real-world gait parameters. Collected data out of the experiments, observations, and interviews make mixed-method research involving quantitative and qualitative sections to answer the research question.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Immersive Virtual Environments (IVEs) are becoming more accessible and more widely utilized in crucial disciplines like training, communication, health care, and education. But supporting natural interaction remains a practical challenge for virtual reality applications within these disciplines. Locomotion in immersive virtual environments is one of the most significant parts of virtual applications. Various locomotion techniques in virtual reality, like controller/joystick locomotion, teleportation, real walking, redirected walking, and gaze-directed locomotion, all provide users with different ways of moving in VR. However, they afford different levels of immersion, ease-of-use, competence, and psychological comfort for VR users [1]. When selecting a locomotion method for a specific application, it is important that the developer consider the specific needs of the application. For instance, requirements for a VR training system are different from a VR rehabilitation system.

Prior work has shown that people walk differently in VR and in the real world; walking with Head Mounted Display(HMD) in the virtual environment gives different gait parameters than when walking in the real world. This seems to be due to both the weight of the HMD and the smaller vertical field-of-view [6]. Also, participants walked slower in the virtual than the physical environment [2]. Researchers evaluated the differentiation of biomechanical walking parameters between real and virtual places and noticed that virtual walking had much lower stability than walking in the real world, that may be explained by some factors like the weight of the HMD [4].

1.1 Research Objectives and Goals

Prior research has established that people's behavior changes when they are in virtual environments, compared to the real world. However, very little study has evaluated how people's perception and action change over time as they gain more experience with VR. My current research project is explicitly focused on users' gait. The

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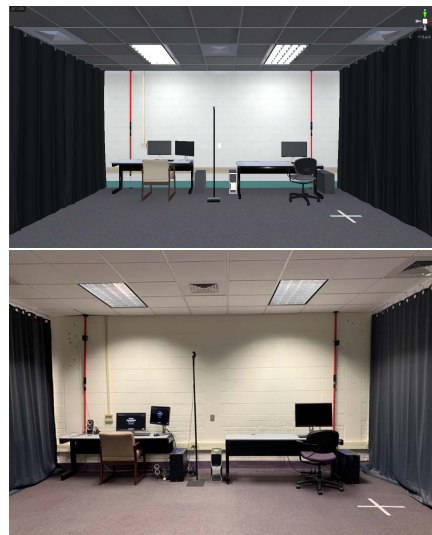


Figure 1: Physical tracked space vs. virtual space in unity

research questions that I explore are 1) What are the differences between walking in the real world compared to walking in Virtual Reality? And 2) How do prior VR experiences make the difference between novice and skilled users walking in VR environments?

1.2 Hypothesis

It is unknown how the different perceptions and actions in the real and virtual world may be affected by experience with VR. We hypothesize that the difference in gait between real and VR may be moderated by a user's experience walking in VR, such that more experienced users' gait will differ less. Therefore, this study provides researchers, developers, and practitioners in virtual reality with interaction-related knowledge concerning gait parameters in VR, so that they can design future models based on strong experimental and analytical expertise.

2 EXPERIMENT DESIGN

We are running a mixed study design with two factors: walking in real or virtual worlds (within-subjects) and level of prior VR experience (between-subjects). We will recruit at least forty participants, 20 novice VR users who have less than 5 hours of experience working with VR applications, and 20 expert VR users, with more than 20 hours of VR experience.

As it is shown in figure 1, we have created a tracked space in a physical lab using the HTC Vive trackers as well as an exact virtual version of it to run the experiments. Participants are supposed to wear feet trackers, a hip tracker, and a head-mounted display(HMD) or a hard hat with a fixed tracker on top of that. Also, they carry one controller in each of their hands to track their hands' positions. They walk back and forth along a 6 meter path between two targets for 40

trials. These experiments are done in both VR and real conditions, wearing the HMD for walking in the virtual room and the hard hat to walk in the real room condition. The order in which the VR and real conditions are presented is counterbalanced.

By running the experiment, we record data of head, hip, hands, and feet positions. Then, the following gait parameters will be extracted from the data: walking speed, head-trunk angle, step count, step length, and total path length. Walking speed is computed as the average speed of a participant while he/she is walking. Head-trunk angle is the angle between the head and the trunk, where 0° is looking perpendicular to the angle of the trunk and positive and negative angles are looking up and down. Step counting requires a step detection algorithm which identifies places that foot is on the ground. By using the position of feet trackers, we can calculate the speed walking; whenever speed is zero, it means that foot is on the ground. Step length is the distance between the point of beginning contact of one foot and the point of beginning contact of the opposite foot.

Linear mixed models are used to analyze the gait parameter data. Modality (VR vs. Real), experience level (novice vs. expert), and gait trial are used as fixed effects. As random effects, participant ID is used for both intercept and slope with respect to modality. P-values will be obtained by likelihood ratio tests of the full model with the effect in question against the model without the effect in question [7]. Then, collected information from post-experiment interviews and observations during running the experiments provides us with qualitative data which are essential to finding out user experience in walking in a virtual environment based on the prior experience [3, 5].



Figure 2: Lab setup for running gait experiments: left setup includes feet trackers, hip tracker, and one tracker fixed on top of hardhat. Right setup is for virtual phase including feet trackers, hip tracker, and HMD with HTC Vive tracker.

3 RESULTS/DISCUSSION

Of the 32 participants, 8 failed to complete the experiment because of the trackers' calibration and data collection issues. 17 novice VR users and 7 expert VR users completed the experiment with no problem. Based on the collected data, I calculate the gait parameters for VR and the real world. So, I found some patterns like a decrease in the walking speed and step length in VR compared to the real world. Besides collecting data on subjects' body movements during walking in the designated place, I could gather valuable information about their perception in the VR environment by observing and interviewing the participants. I will analyze all the interviews, observation notes, and recorded videos to determine how prior experience affects people's perception of the virtual environment.

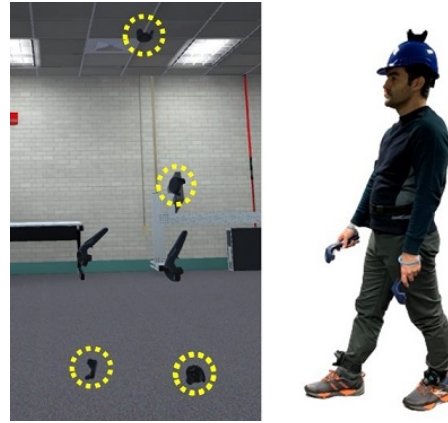


Figure 3: HTC Vive trackers on user's body

4 CHALLENGES/FEEDBACK

In this line of research, I'd like to receive feedback and suggestions on conducting this series of study. One of my concerns is the prior VR experience; in these experiments, participants as novice or skilled VR users, walk between two targets on the ground. Users' previous experience helps us divide them into two groups of novice and experienced users. However, they may gather experience by using some stationary VR applications and platforms. So, the expert VR users may have gained a lot of experience in the virtual environments without walking in the VR. What are the ways to find meaningful influences of prior experience on people's perception and action in VR? Another area of concern is the experiment conditions; we designed the experiment focused on users walking. During the 80 trials, users only walk on the same path with or without HMD. Can we use these results for other VR applications that require users to do a specific task while walking? These questions are a few topics that I would appreciate feedback on.

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REFERENCES

- [1] C. Boletsis and J. E. Cedergren. VR Locomotion in the New Era of Virtual Reality: An Empirical Comparison of Prevalent Techniques. *Advances in Human-Computer Interaction*, 2019, 2019. doi: 10.1155/2019/7420781
- [2] P. W. Fink, P. S. Foo, and W. H. Warren. Obstacle Avoidance During Walking in Real and Virtual Environments. *ACM Transactions on Applied Perception*, 4(1):2, 2007. doi: 10.1145/1227134.1227136
- [3] P. Gill, K. Stewart, E. Treasure, and B. Chadwick. Methods of data collection in qualitative research: Interviews and focus groups. *British Dental Journal*, 204(6):291–295, 2008. doi: 10.1038/bdj.2008.192
- [4] O. Janeh, E. Langbehn, F. Steinicke, G. Bruder, A. Gulberti, and M. Poetter-Nerger. Walking in virtual reality: Effects of manipulated visual self-motion on walking biomechanics. *ACM Transactions on Applied Perception*, 14(2):1–15, 2017. doi: 10.1145/3022731
- [5] C. McGrath, P. J. Palmgren, and M. Liljedahl. Twelve tips for conducting qualitative research interviews. *Medical Teacher*, 41(9):1002–1006, 2019. doi: 10.1080/0142159X.2018.1497149
- [6] B. Mohler, J. Campos, M. Weyel, and H. H. Bülthoff. Gait parameters while walking in a head-mounted display virtual environment and the real world. In *13th Eurographics Symposium on Virtual Environments and 10th Immersive Projection Technology Workshop (IPT-EGVE 2007)*, pp. 85–88. Eurographics Association, 2007.
- [7] B. Winter. Linear models and linear mixed effects models in R with linguistic applications. *arXiv preprint*, arXiv:1308, 2013.